DC Motor Angular Speed Control

Eric Castro Velásquez*, Esteban Segura Garcia*, Daniel Chavarria Garcia*

*School of Electronics Engineering, Instituto Tecnológico de Costa Rica (ITCR), 30101 Cartago, Costa Rica
{ecastro, essegura, dachavarria}@estudiantec.cr

***Abstract*—This project presents the design, simulation, and experimental validation of a proportional–integral (PI) controller for the angular speed of a DC motor. The plant was identified through repeated measurements and statistical analysis. The design of the controller was based on root-locus techniques, to satisfy the design specifications. Implementation was in both domains, continuous and discrete. The simulation results obtained in Simulink show that the design meets the specifications. The best controller obtained a settling time of 265 ms and an overshoot of 0.49% in simulation, while maintaining acceptable performance experimentally. The experimental implementation shows discrepancies due to model limitations, particularly in the discrete domain. Additionally, distortions in the physical plant response and measurement noise affect system performance. For this reason, this work studies the impact of noise on the results and its influence on the overall system behavior.**

***Keywords*—DC motor, discrete-time control, overshoot, PI controller, root locus.**

I. Introduction

The control of angular velocity in DC motors is a classic automatic control problem with broad industrial applications. A standard solution is the proportional-integral (PI) regulator, whose integral action guarantees zero steady-state error for step inputs and provides disturbance rejection capability [1].

In this work, a PI regulator is designed for the *hps5130* DC motor system, modeled as a first-order plant identified experimentally. The design is carried out in both continuous and discrete time with a sampling period of $T_s = 5\,\mathrm{ms}$, targeting a 2% settling time below $300\,\mathrm{ms}$ and overshoot below 3%. Simulation results in *Simulink* and experimental validation on the *kitprog059* platform are reported [3].

II. Theoretical Background

A. DC Motor as a Control Plant

The DC motor angular speed is modeled as a first-order transfer function:

$$G(s) = \frac{K}{s+p}, \tag{1}$$

where K is the static gain and p is the dominant pole, both obtained through experimental identification [2].

B. PI Regulator

The continuous-time PI controller is expressed as:

$$PI(s) = K_P + \frac{K_I}{s} = \frac{K_c(s-z_0)}{s}, \tag{2}$$

where K_P sets the response speed, K_I eliminates steady-state error, and z_0 is the controller zero whose position relative to the plant pole shapes the closed-loop dynamics [2].

C. Block Diagram

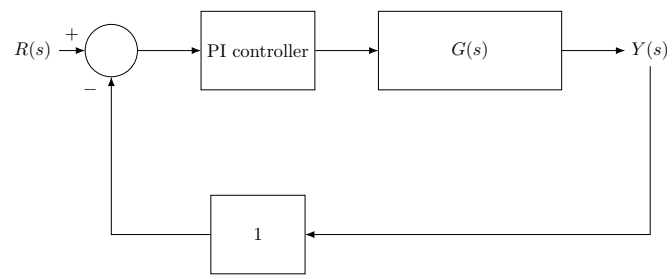

Figure 1. Closed loop system with PI Controller.

D. Root-Locus Design

The closed-loop poles satisfy $1 + K_c \cdot G_{OL}(s) = 0$, where $G_{OL}(s)$ is the open-loop transfer function. Design specifications impose the following constraints on the dominant pole location [2]:

$$\sigma = \frac{4}{t_s}, \qquad \zeta = \frac{-\ln(M_p/100)}{\sqrt{\pi^2 + \ln^2(M_p/100)}}, \tag{3}$$

where σ is the minimum required real part, t_s is the 2% settling time, and ζ is the damping ratio corresponding to the maximum overshoot M_p.

E. Controller Discretization

The continuous PI is discretized via the backward Euler approximation $s \approx (z-1)/(T_s z)$, yielding:

$$PI(z) = \frac{b_1 z + b_0}{z + a_0}, \tag{4}$$

where coefficients b_1, b_0, and a_0 depend on K_P, K_I, and T_s. The chosen $T_s = 5\,\mathrm{ms}$ provides between 30 and 75 samples within the settling time, ensuring adequate dynamic representation [1], [2].

III. Methodology

A. Plant Model Selection

Three CSV datasets were collected from the *hps5130* motor under the same operating conditions. A statistical analysis (mean, standard deviation, and coefficient of variation) determined the most representative dataset, which was then used to identify an empirical first-order model via the *System Identification* toolbox in MATLAB, following (1). The RMSE computed over the remaining datasets served as the generalization criterion for model selection.

B. PI Controller Design

Three PI controllers were designed in continuous and discrete time using the *sisotool* root locus editor in MATLAB. Each design placed the controller zero z_0 at a different location relative to the dominant plant pole, yielding three distinct dynamic responses. All designs were constrained to $t_s < 300\,\text{ms}$ and $M_p < 3\,\%$, as imposed by (3). Discrete equivalents were obtained via `c2d` at $T_s = 5\,\text{ms}$ and verified in *Simulink* under step and disturbance inputs.

C. Experimental Validation

The best-performing discrete-time controller was implemented on the *kitprog059* platform. Step reference signals and input disturbances were applied to evaluate settling time, overshoot, and steady-state error under real operating conditions.

IV. Results

A. Model Selection Results

As a result of the statistical analysis, the model obtained from Rep. 3 was selected as the best representation of the plant. The identified transfer function is given by:

$$G(s) = \frac{12.535}{s + 9.165} \tag{5}$$

This model achieved an average fit of approximately 90% based on the experimental data.

B. PI Controller results

According to the plant model, the continuous PI parameter for the selected controller are displayed in table I.

Table I
Continuous-time PI controller parameters.

PI Controller	K_c	z_0
PI1	1.0707	-9.4

$$PI(s) = \frac{1.0707 \cdot (s + 9.4)}{s} \tag{6}$$

The discrete PI parameter for the controller are displayed in table II.

Table II
Discrete-time PI controller parameters.

PI Controller	b_1	b_0	a_0
PI1	1.222	1.178	-1

$$PI(z) = \frac{1.222z + 1.178}{z - 1} \tag{7}$$

C. Continuous-Time Simulation Results

Figure 2 shows the step response of the third continuous-time PI controllers. PI met the specifications, achieving settling times of $310\,\text{ms}$ and $391\,\text{ms}$ respectively, both with zero overshoot.

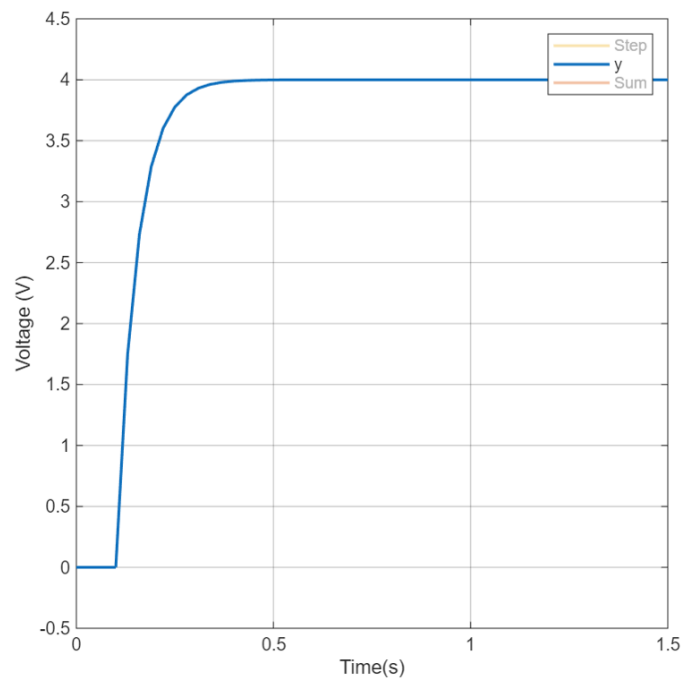

Figure 2. Step response of the three continuous-time PI controllers in Simulink.

D. Discrete-Time Simulation Results

Figure 3 presents the step response of the best discrete-time controller (PI_1), which achieved $t_s = 265\,\text{ms}$ and $M_p = 0.49\%$, both within specifications.

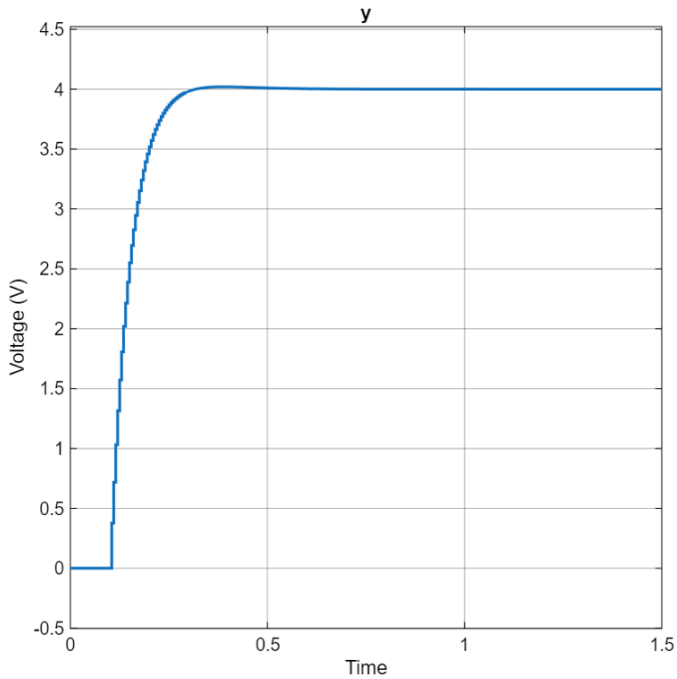

Figure 3. Step response of the best discrete-time PI controller in Simulink.

E. Experimental Results

Figure 4 shows the experimental step response of the continuous-time PI controllers implemented on the *kitprog059* platform.

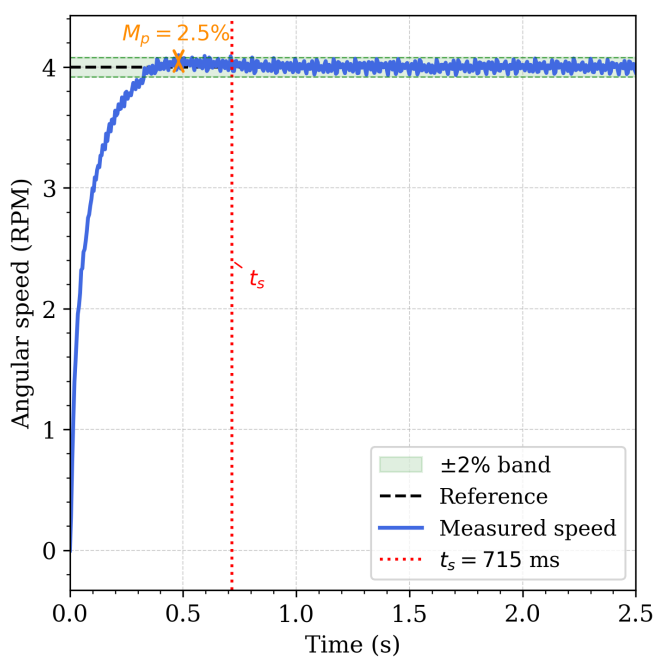

Figure 4. Experimental step response of the PI controllers on the *kitprog059* platform (continuous-time implementation).

Figure 5 presents the corresponding experimental step response for the discrete-time implementation.

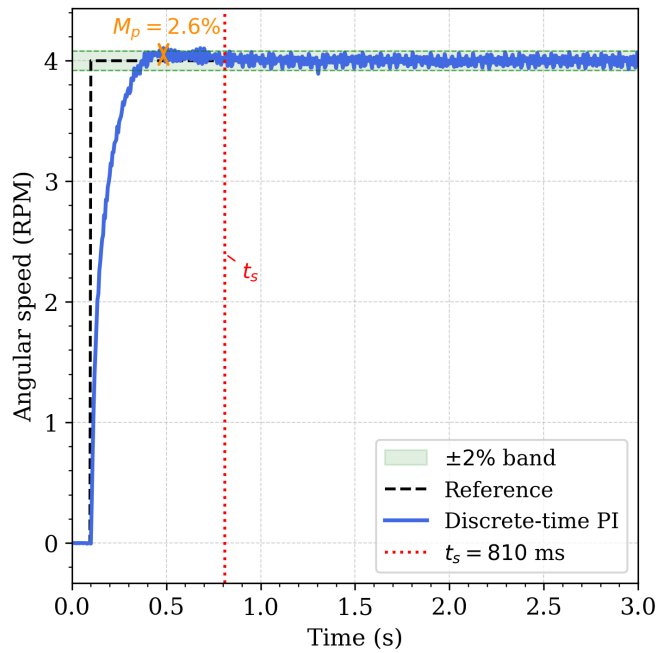

Figure 5. Experimental step response of the PI controllers on the *kitprog059* platform (discrete-time implementation).

F. Performance Summary

Tables III and IV summarize the performance metrics for simulation and experimental results respectively. Values of IV have being obtained by the script in TECDigital.

Table III
SIMULATION PERFORMANCE METRICS OF THE DESIGNED PI CONTROLLERS.

Controller	**Domain**	t_s **(ms)**	M_p **(%)**	e_{ss}
PI_1	Continuous	310	0	0
PI_2	Continuous	391	0	0
PI_3	Continuous			
PI_1	Discrete	265	0.49	0
PI_2	Discrete	370	0	0
PI_3	Discrete	370		

Table IV
EXPERIMENTAL PERFORMANCE METRICS OF THE DESIGNED PI CONTROLLERS.

Controller	**Domain**	t_s **(ms)**	M_p **(%)**	e_{ss}
PI_1	Continuous	330	0.95	±0.0235
PI_2	Continuous	315	0.83	±0.03
PI_3	Continuous	330	1.26	±0.0313
PI_1	Discrete	165	23.4	±0.034
PI_2	Discrete	180	2	±0.032
PI_3	Discrete	230	1.45	±0.0305

V. Discussion

A. Statistical Analysis of Plant Identification

Tables V and VI summarize the parameter estimation results. For the velocity model, the gain K shows excellent repeatability (CV = 1.64%), while the time constant τ presents high variability (CV = 47.14%), likely caused by noise in the transient response and variability in initial motor conditions. For the current model, the situation reverses: τ is highly repeatable (CV = 4.61%) whereas K shows moderate dispersion (CV = 13.93%).

Table V
Experimental results for velocity model identification.

Parameter	Rep. 1	Rep. 2	Rep. 3	Statistics
K	1.3342	1.3708	1.3746	$\mu = 1.3598$, $s = 0.0223$, $CV = 1.64\%$
τ (s)	0.2301	0.1099	0.1071	$\mu = 0.1490$, $s = 0.0703$, $CV = 47.14\%$

Table VI
Experimental results for current model identification.

Parameter	Rep. 1	Rep. 2	Rep. 3	Statistics
K	0.1644	0.1326	0.1282	$\mu = 0.1418$, $s = 0.0197$, $CV = 13.93\%$
τ (s)	0.1090	0.1081	0.1174	$\mu = 0.1115$, $s = 0.0051$, $CV = 4.61\%$

Table VII shows that the model from Rep. 3 achieves the lowest validation RMSE (0.2113), making it the best compromise between fit accuracy and generalization. The Rep. 1 model, despite the lowest training error (0.0952), exhibits the highest validation RMSE (0.4089), indicating overfitting.

Table VII
Training and validation RMSE for the identified transfer functions.

Training	Validation	RMSE (fit)	RMSE (val.)
Rep. 1	Rep. 2, 3	0.0952	0.4089
Rep. 2	Rep. 1, 3	0.1122	0.2220
Rep. 3	Rep. 1, 2	0.1114	0.2113

B. Noise Analysis

The angular speed signal was analyzed during closed-loop operation under a step input with disturbance, sampled at $100\,\mathrm{Hz}$ over a total duration of $17.87\,\mathrm{s}$ (1787 samples). Figure 6 shows the time-domain signal along with its power spectral density (PSD).

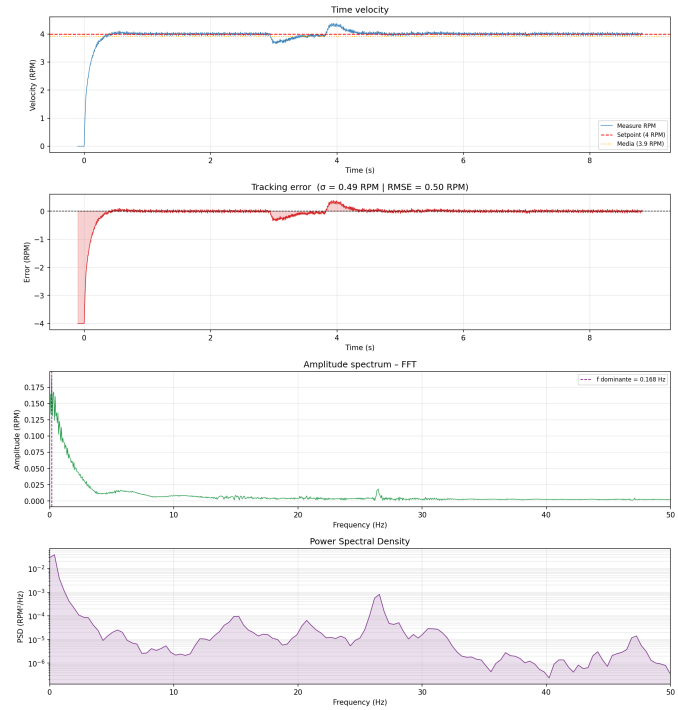

Figure 6. Noise Analysis.

The signal mean was $3920\,\mathrm{RPM}$ against a setpoint of $4000\,\mathrm{RPM}$, yielding a static error of only $-80\,\mathrm{RPM}$ (-2%), which confirms the integral action effectively eliminates steady-state error in practice. The standard deviation was $\sigma = 490\,\mathrm{RPM}$ and the peak-to-peak amplitude reached $4390\,\mathrm{RPM}$, representing 12.3% of the reference speed. The RMSE with respect to the setpoint was $500\,\mathrm{RPM}$, consistent with the σ value given the near-zero mean error.

Spectral analysis revealed a dominant frequency component at $0.168\,\mathrm{Hz}$ with an amplitude of $190\,\mathrm{RPM}$, suggesting the disturbance is not purely random but contains a low-frequency oscillatory component, likely associated with mechanical load periodicity or commutator switching. The total noise power estimated from the PSD was $2 \times 10^4\,\mathrm{RPM}^2$, yielding a signal-to-noise ratio of $18.04\,\mathrm{dB}$. This moderate SNR explains the high coefficient of variation observed for τ in velocity model identification (CV = 47.14%): the transient slope estimation is sensitive to the low-frequency component at $0.168\,\mathrm{Hz}$, which distorts the apparent settling behavior and inflates the variance of τ across repetitions. The current model, whose transient is faster and less affected by this component, showed a much lower variability (CV = 4.61% for τ). These results suggest that a pre-identification low-pass filter or signal averaging could reduce τ uncertainty in future experiments.

C. Experimental versus Simulation Results

The experimental results reveal discrepancies with respect to simulation predictions, as expected from a first-order model that does not capture friction, dead zone, or amplifier nonlinearities.

For the continuous-time controllers, the experimental settling times were $330\,\mathrm{ms}$, $315\,\mathrm{ms}$, and $330\,\mathrm{ms}$ for PI_1, PI_2, and PI_3 respectively, compared to simulated values of $310\,\mathrm{ms}$ and $391\,\mathrm{ms}$ for PI_1 and PI_2. The deviations are moderate and within the same order of magnitude, indicating that the identified first-order model captures the dominant dynamics adequately. All three continuous-time controllers remained within the $3\,\%$ overshoot specification experimentally (0.95%, 0.83%, and 1.26% for PI_1, PI_2, and PI_3 respectively), confirming the validity of the root-locus design for this implementation domain. The non-zero steady-state error observed ($e_{ss} \approx \pm 0.02$–0.03) is attributed to sensor quantization and measurement noise at the operating speed, rather than a structural failure of the integral action.

The discrete-time controllers showed a more pronounced divergence from simulation. PI_2 and PI_3 met the overshoot specification with 2.0% and 1.45% respectively, and achieved settling times of $180\,\mathrm{ms}$ and $230\,\mathrm{ms}$, both below the simulated values, suggesting that the real plant dynamics are slightly faster than the identified model at this operating point. In contrast, PI_1 exhibited an overshoot of 23.4%, far exceeding the $3\,\%$ specification despite meeting it in simulation ($0.49\,\%$). This anomalous behavior is consistent with the zero placement of PI_1 being closest to the plant pole, which makes the closed-loop response more sensitive to model mismatch and to the quantization effects inherent in discrete implementation at $T_s = 5\,\mathrm{ms}$. The interaction between the low-frequency noise component identified at $0.168\,\mathrm{Hz}$ and the aggressive gain of PI_1 likely amplified the transient overshoot under real conditions. PI_2 therefore emerges as the best overall discrete-time design, satisfying all specifications in both simulation and experiment with the best generalization to the real plant.

Additionally, a discrepancy was observed when comparing the performance metrics reported in Tables III and IV with those obtained from the graphical analysis using a Python-based script. In particular, for the continuous-time response shown in Figure 4, the settling time was approximately $715\,\mathrm{ms}$ with an overshoot of 2.5%, which, although still within the specified limits, differs significantly from the values reported using the TECDigital script. A similar situation was observed in Figure 5, where the settling time was approximately $810\,\mathrm{ms}$ and the overshoot reached 2.6%.

In both cases, the steady-state error remained within the required bounds, indicating that the integral action was still effective. However, these differences suggest that the TECDigital script may produce inaccurate estimates of settling time and overshoot, and therefore may not fully reflect the actual dynamic response of the controller. This highlights the importance of validating automated performance metrics with direct analysis of the system response.

VI. Conclusions

This work demonstrates the effectiveness of a PI controller for DC motor speed regulation. The plant was identified as a first-order system using experimental data, and the statistical analysis provided an adequate representation of the system dynamics. By applying pole cancellation and root-locus techniques, the closed-loop system met the specified performance requirements in simulation, including settling time and overshoot.

Nevertheless, experimental validation revealed discrepancies caused by model limitations and unmodeled effects in the physical system, such as noise. These findings underscore the importance of considering real-world conditions during the controller design process. As a first attempt at controlling a real plant, the model proved useful, but it also highlighted the gap between simulation and practice. In particular, noise and modeling uncertainties must be addressed when designing control systems for practical applications like DC motor angular velocity regulation. Ignoring these factors risks producing controllers that perform well in simulation but fail to deliver the desired behavior in real implementations.

Therefore, future work should explore robust controller design, incorporate signal filtering, and improve system modeling to enhance performance and reliability. Beyond the technical aspects, this project also served as a valuable learning experience, showing how theory and practice often diverge and reminding us that engineering solutions must adapt to the imperfections of real-world systems.

References

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